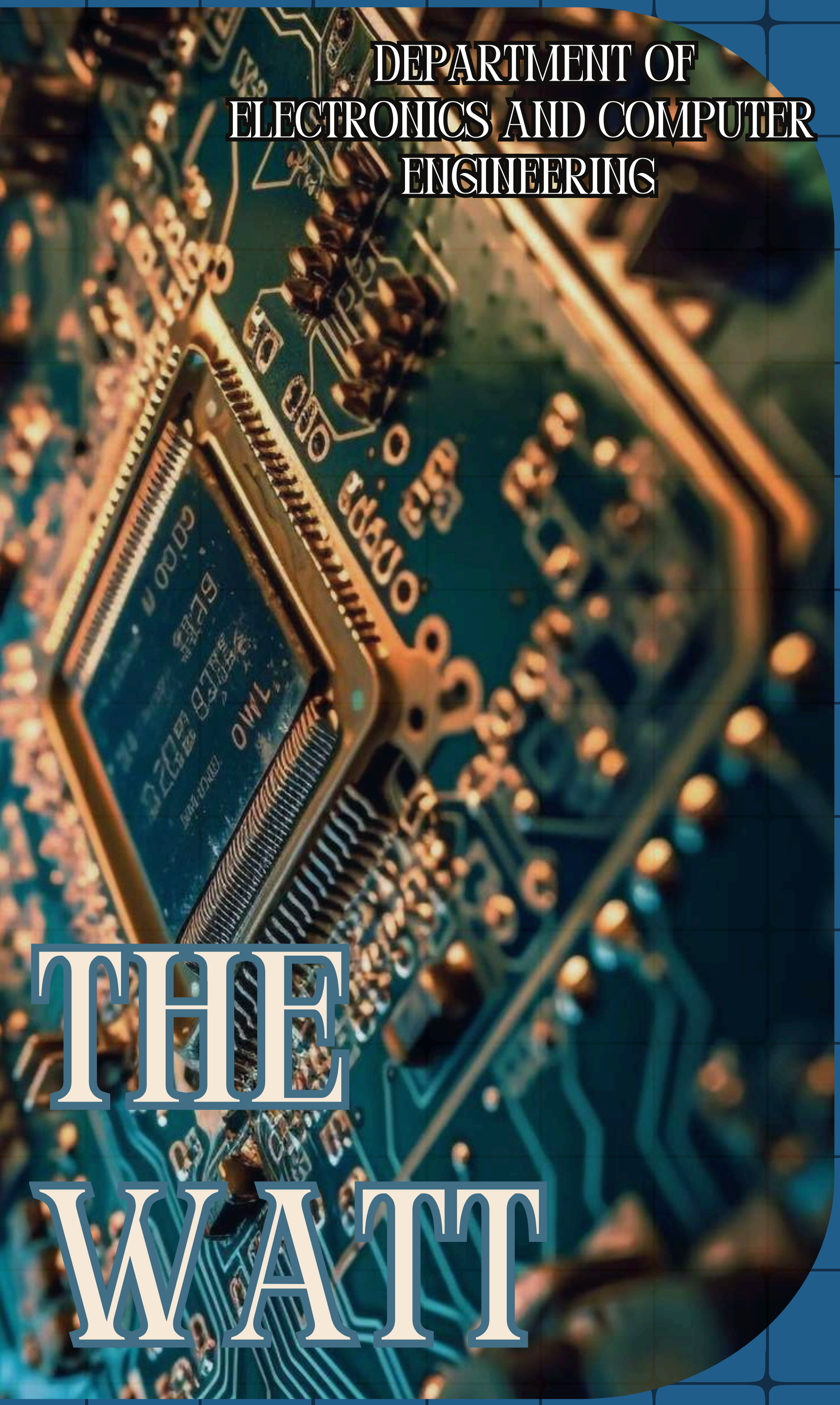
FR. AGNEL POLYTECHNIC, VASHI

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING



THE WATT

ACADEMIC YEAR
2023-24

ELECTRONICS AND COMPUTER ENGINEERING

VISION

Be a premier Electronics and Computer Engineering Department creating competent diploma engineers, focusing on holistic development with Indian values and ethics.

MISSION

- To impart industry oriented quality education and training in the field of Electronics and Computer Engineering.
- To provide value based education and opportunities for holistic development.
- To inculcate awareness towards societal needs and environmental issues.

Electronics and Computer Engineering

Program Educational Objectives (PEOs)

1. Computer Software and Hardware Usage
Use state-of-the-art technologies for operation and application of computer software and hardware.

2. Electronics and Computer Systems
Maintain various types of Electronics and Computer systems

PROGRAM SPECIFIC OUTCOMES (PSOs)

1. Provide socially responsible, environment friendly solutions to Electronics and Computer engineering related broad-based problems adapting professional ethics.

2. Adapt state-of-the-art Electronics and Computer engineering broad-based technologies to work in multi-disciplinary work environments.

3. Solve broad-based problems individually and as a team member communicating effectively in the world of work

TABLE OF CONTENTS

- Rising Stars
- Zest 2024
- Technocratz 2024
- Legacy makers
- technical paper presentation
- Expert lectures and seminars

RISING STARS

- ARTICLES
- STORIES
- POEMS
- ARTWORK



ARTICLES

LOSING A MOTHER

Losing a mother is one of the most devastating things in the world. It happens and we somehow find the strength to go on. The days turn into weeks, the weeks turn into months, and before we know it here we are years later, wondering how we made it this far without her physically in our lives.

You will always miss your mom but there are moments in life when you miss her a little more than usual. You will miss her when you are simply in the store shopping and you see the other childrens shopping with their mothers. You will miss her when something great happens and she is the first person you want to call but you can't. You will miss her on mother's day when everyone else is celebrating and you feel alone. You will miss her when you had a bad day.

Losing the women who loved you unconditionally is a pain that cannot be described in words. It is virtually impossible to move on from the memory of losing the women who sacrificed happiness in her own life so that you could have a better one. A mother's love is truly irreplaceable.

Sometimes I wish for my mom to come back but I don't want her to suffer again. I know she is with me and I will always love and miss her with all my heart.

Somedays I wish I could at least go back and relive days with her. Not to change anything, but to feel few things twice. I miss you mumma"

~ Neerah Khan
TE 3K



KEYS TO SUCCESS

There are countless paths to success, but in my opinion, the most successful individuals are those who are hard-working, dedicated, and focused on their goals. If you want to move forward in life, then work diligently every single day. You may not succeed immediately, but if you continue to work on yourself, the day may come when you do succeed. Success is not reserved for those who are destined, but for those who are determined. Many people only see the glamorous side of success and believe they can attain it, but when they actually attempt it, only a few succeed, as the rest give up when they encounter challenges. Success is the ability to do what you want, when you want, where you want, with whom you want, as much as you want.

In the midst of all the busyness, it's crucial to take time away from the pressure to regroup and recharge so you can be more effective. Nonstop worrying burns people out. Don't stress about what's going on in your head. Let your heart make it all okay. Every minute you spend being upset, worried, or anxious is a lost opportunity to be happy, relaxed, and content. Instead of stressing past the moments, enjoy them. Believe in yourself and your abilities.Embrace the journey, even when the road gets tough. It's in the struggle that you find your true strength and resilience.

~Shaan Jamabavlikar
TE 3K

Music: The Essence of Life

Have you ever taken just five minutes out of your hectic day to ponder that everything in nature seems to dance to a melody that we cannot hear? It's as if nature itself moves to a rhythm beyond our understanding. Why is this so?

The answer is simple: it's an invitation to immerse yourself in the melodies played by nature itself. In our modern world, we often find ourselves caught in a relentless rat race, pursuing materialistic goals that we cannot take with us when we die. The only escape from this constant loop is through music. Music offers a solution to many of life's problems. Yet, in our busy lives, we often overlook the melodies around us—sounds that have the power to free us from stress, anxiety, and the daily grind. How often do we truly stop and listen to the healing and inspiring sounds that can reconnect us with our deepest selves?

The music produced today often falls short of genuine artistry. We cannot simply combine random beats and lyrics and call it music. True music is crafted with skill and technique, a divine gift bestowed upon a few. Sadly, the number of true artists is dwindling. Today, many create music merely for profit, aiming to catch attention rather than to create something meaningful. This shift has led to a generation that often fails to recognize the depth of genuine music.

Ludwig van Beethoven, one of the greatest composers and pianists of his time, who continued to create masterpieces even after losing his hearing, once said, "Music is the mediator between the spiritual and the sensual life."

This quote emphasizes how music connects our inner, spiritual experiences with the external, physical world. It harmonizes our deepest emotions with tangible sensations, transforming mere sound into something that touches the most profound aspects of human existence.

Consider the evening breeze that reminds you of your vitality, the melodious chirps of crickets, the harmonious songs of birds at dawn, or the rhythmic drumming of woodpeckers. These are all forms of music that can inspire and uplift us. Similarly, music created by god-gifted artists enriches mankind, offering solace and inspiration through their profound talent.

As the saying goes, "Where words fail, music speaks." This reflects the deep importance of music in our lives. Let music envelop you completely, allowing it to redeem and transform you. Jim Morrison, the enigmatic lead singer of The Doors, once remarked, "Music inflames temperament." This insight reveals how music stirs and intensifies our emotions, amplifying our feelings—whether joy, anger, sadness, or excitement.

In conclusion, music holds a power beyond human comprehension. I encourage you to move beyond the superficial sounds often labelled as music and seek out the genuine compositions that can transform and elevate your life.

~ Darshan Bornare
TE 51

TECHNICAL ARTICLE

SMART TOLL BOOTH SYSTEM

The current toll collection system in India used RFID technology, where all vehicleswere charged the same toll regardless of how far they travelled. This created problems for people living near toll booths, as they had to pay the same amount as those travelinglong distances. Althoughthe government had introduced rules to exempt people living within 5-6 km from toll charges, these rules were not always followed properly.

The paper presented a solution: a SMART TOLL BOOTH SYSTEM that utilized GPS technology to calculate tolls based on the distancea vehicle travelled. By assigning GPS coordinates to specific areas around toll booths, the system ensured that local travellers paid a fair toll accordingto the actual distance they drove. The system combined RFID for vehicle identification with GPS for tracking journeys,allowing the toll to be deducted automatically based on the distancetravelled.

This system aimed to make toll collection fairer, especially for residents living nearby to the toll booth, and improvethe efficiency of the tolling process. It could be easily scaled and applied across the country to ensure that all drivers paid a fair toll based on their journey.



Rohan Pednekar (TE 61)



Stesha Parkhe (TE 61)

BIOINFORMATICS COMPUTING

Verbal communication is vital for humans, but certain individuals are unable to speak due to specific conditions. There are two categories of mute persons: those who cannot talk because they are unable to hear, and those who suffer from articulation limitations due to brain disorders such as paralysis or autism. The latter can hear and understand speech but cannot articulate it. This group needs a system that can decode their imagined speech - speech they comprehend but cannot express. The ultimate goal is to develop a wearable, daily-use system that will noninvasively acquire electrical signals from the speech areas in the brain.

Various modalities, both invasive and non-invasive, are used to decode imagined speech, including Electrocorticography (ECoG), Intracortical electroencephalography (ICE), functional Magnetic Resonance Imaging (fMRI), Positron Emission Tomography (PET), functional Near Infrared (fNIR), Electroencephalography (EEG), and Magnetoencephalography (MEG). Each of these modalities has its own limitations. Given the requirements of portability, wearability, and user comfort, EEG and MEG are preferred for this research. However, overcoming their inherent drawbacks - low spatial resolution in EEG, poor temporal resolution in MEG, and low signal-to-noise ratios in both – is a challenge. High spatial and temporal resolution is necessary to enhance feature extraction and recognition. Our research focused on selecting the appropriate electrodes, locating the exact speech areas the scalp, and improving signal processing and analysis methods to address these challenges and advance brain-computer interfaces (BCIs).

When neurons in the brain are activated, they generate electrical and magnetic signals. Identifying the locations where these signals arise during imagined speech requires physiological and anatomical studies. Research has identified two key areas in the brain involved in speech production: the inferior frontal gyrus (Broca's Area) and the temporal superior gyrus (Wernicke's Area). These areas are responsible for articulation and comprehension, respectively.

In our research we have identified the correct speech area on the scalp, reduced the number of electrodes needed and minimized bandwidth contamination. Through a detailed study of EEG electrodes we developed graphene-Ag/AgCl and graphene-graphite electrodes. These electrodes were validated against the standard Ag/AgCl wet electrodes, showing satisfactory performance. Spectrotemporal parameters were then used to classify six Marathi phonemes using SVM, Random Forest, and DNN algorithms. Our research achieved comparable accuracy and F1 scores to previous studies but with only three electrodes, inclue electrode. Our work advances the field of brain-computer interfaces by decoding imagined sp from brain signals with non-invasive techniques.

Individuals facing verbal communication impairments due to brain disorders, such as paralysis or autism, struggle to articulate speech despite having intact cognitive processes. This paper presents an AI-driven, machine learning-based system that decodes imagined speech using electroencephalogram (EEG) signals. By employing advanced algorithms like Support Vector Machines (SVM) and Deep Neural Networks (DNN), the system processes EEG data to classify imagined phonemes. Key features, such as spectrograms and spectral entropy, are extracted to train these machine learning models. The aim is to provide a non-invasive, wearable device that enables individuals with speech impairments to communicate via their thought processes. Our approach, rooted in artificial intelligence (AI) and machine learning (ML), offers high accuracy in decoding EEG signals into speech, enhancing social participation and overall well-being.



Sejal Jogale (TE 61)



Tanaya Deshmukh(TE 61)

EVENT MEMORY SYSTEM

The IoT-Based Event Memory System (IEMS) is a technology-driven solution designed to enhance event photography by integrating Raspberry Pi, AI, cloud computing, and IoT. It aims to revolutionize the way event photography is managed, providing an automated and real-time solution for capturing, organizing, and sharing memories efficiently. Traditional event photography often suffers from inefficiencies such as delayed photo access, lack of personalization, and high workload for photographers. IEMS overcomes these challenges by offering a seamless, intelligent, and highly interactive system.

The system enables real-time photo uploads to the cloud, where AI-powered face recognition algorithms automatically sort and categorize images into personalized galleries. This allows guests to instantly access their photos through interactive kiosks, mobile apps, and secure web platforms. The smart photo booths powered by Raspberry Pi further enhance the experience by allowing guests to take and customize their pictures with event-themed filters before uploading them to the cloud.

Security and privacy are key considerations, with encrypted cloud storage and controlled access to personalized galleries. The scalability of IEMS makes it adaptable for events of all sizes, from small gatherings to large-scale festivals. By integrating IoT and AI, IEMS provides a seamless, user-friendly, and highly personalized way to capture and relive event memories, setting a new benchmark for event photography technology.



Aayush Bhat
(TE-4K)



Soham Dhanokar
(AN-4K)



Parth Naukudkar
(TE-4K)

POEMS

THE BLOSSOMED FLOWER

I have worked with your flower,
And helped it to grow.
I'm returning it now,
But I want you to know;
This flower is precious,
As dear as can be,
Love it take care of it.
And you will see;
A bright new bloom everyday.
It grew and blossomed;
In such a wonderful way.
Before it was a bud,
Now a lovely blossom.
Im returning it to you,
Remember this flower,
As dear as can be,
Though rightfully yours,
Its part will always belong to me.

~Aayusha Mhatre
TE3k

IN LOVING MEMORY OF MY BEAUTIFUL MOTHER

God saw you getting tired
and a cure was not to be.
So He put His arms around you
and He whispered "Come to me"
With tearful eyes we watched you.
We watched you face away.
Although we love you dearly,
we could not make you stay.
A golden heart stopped beating ,
hard working hands at rest.
God broke our hearts to prove to us,
he chose to take the best.
It's lonesome here without you.
we miss you more each day.
Life doesn't seem the same
since you have gone away.
When days are sad and lonely
and everything goes wrong,
we seem to hear you whisper
"Cheer up and carry on"
Each time we see your picture
you seem to smile and say
"Don't cry, I am in god's hands,
we'll meet again someday!"

~ Neerah Khan
TE 3K



ARTWORK

Resin Art









~ Aayusha Mhatre
TE 4k

Sketches



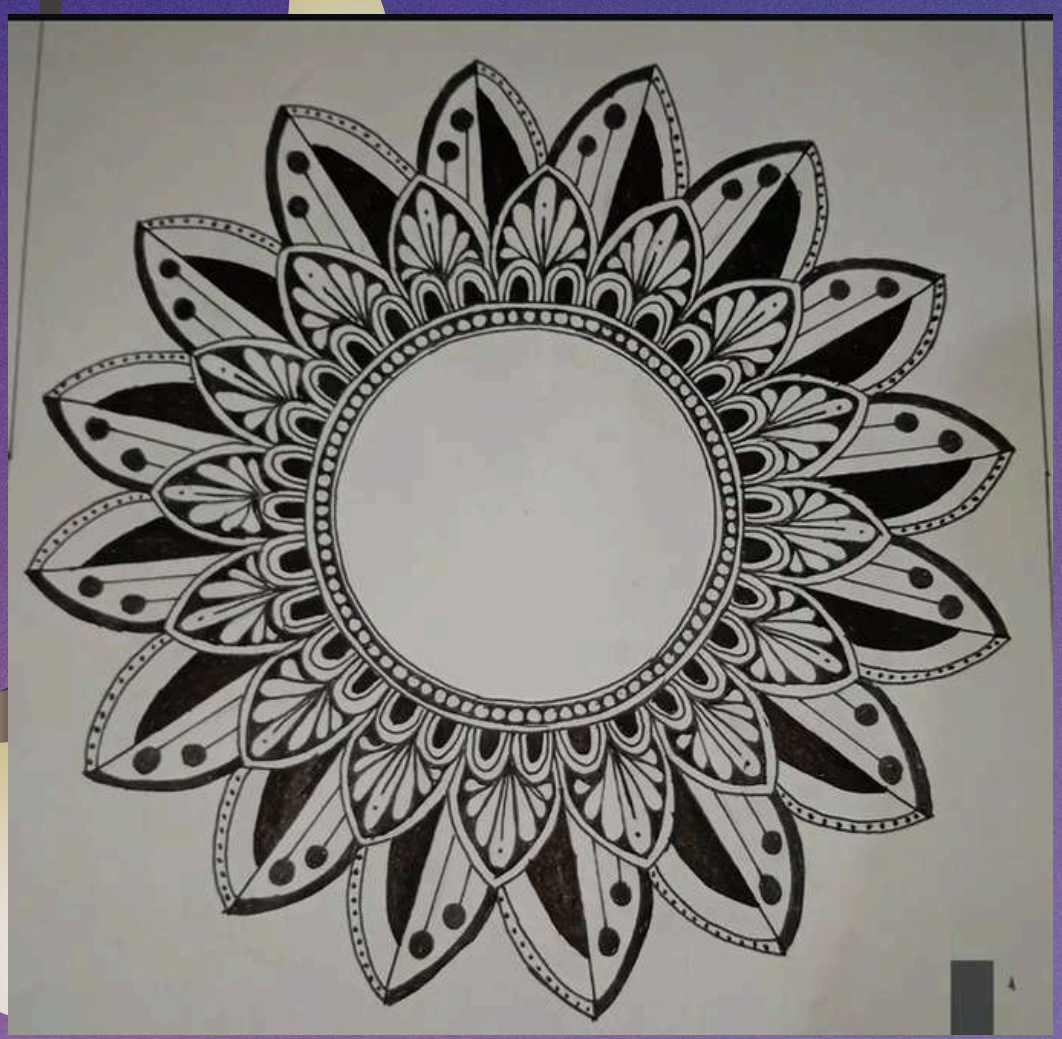




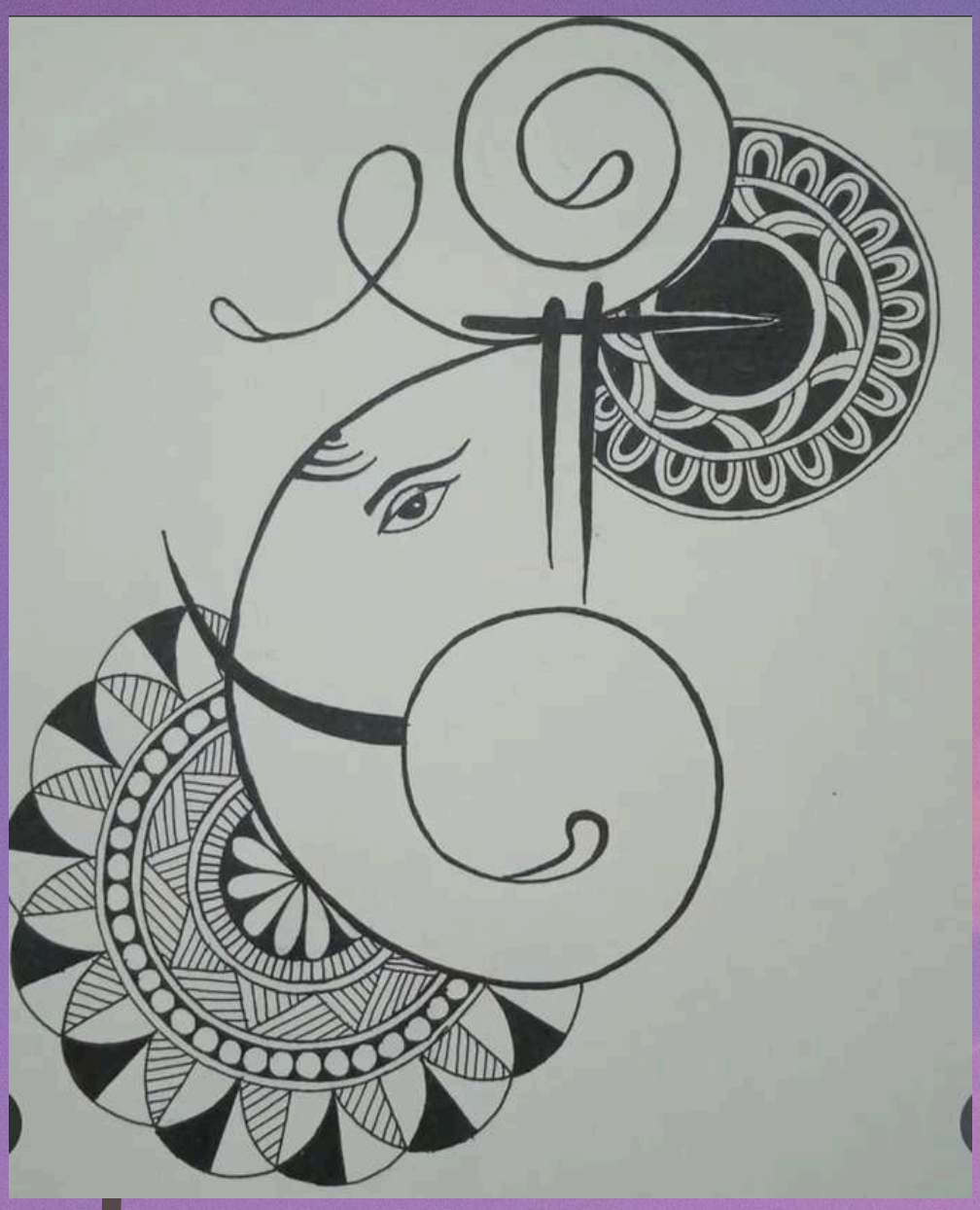


~ *Aarav Girish*
TE 51

Mandala Art











~ Aayusha Mhatre
TE 4k

Sketches & Embroidery Art







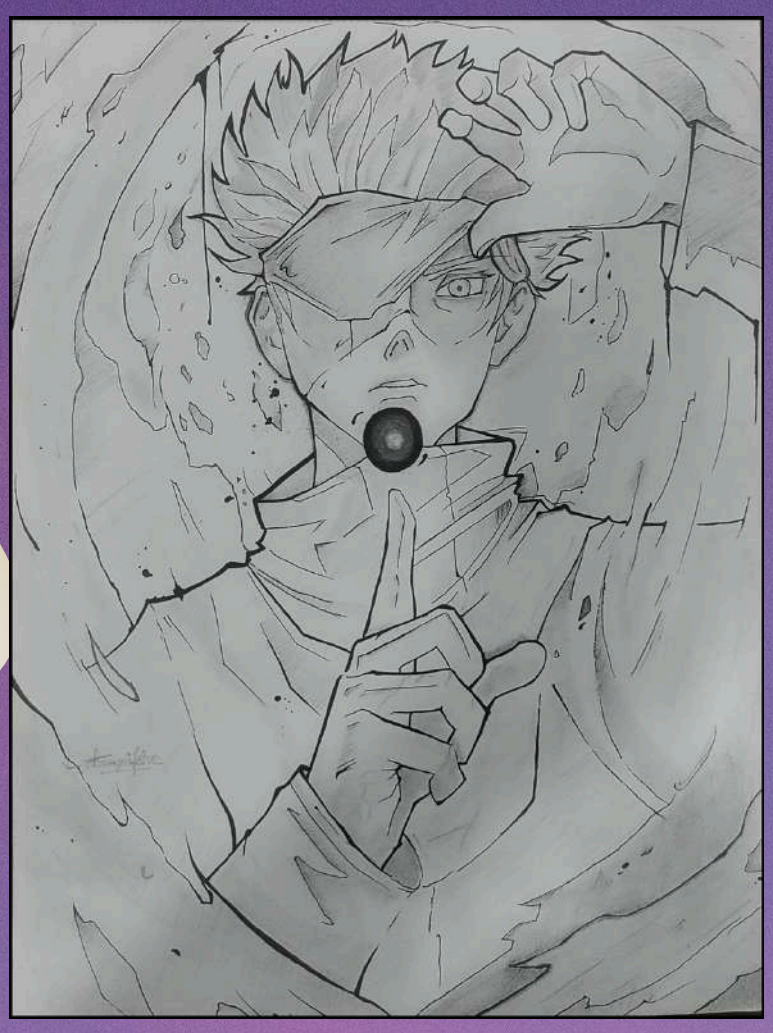


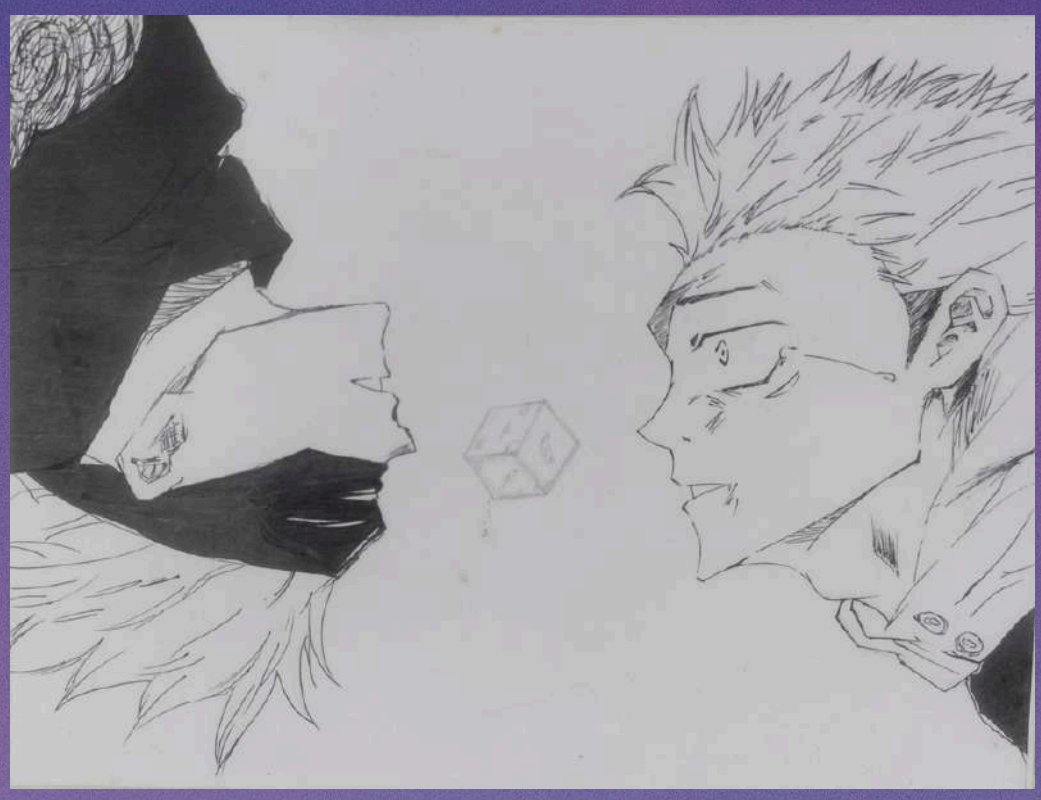
~ Nidhi Koli
TE 51

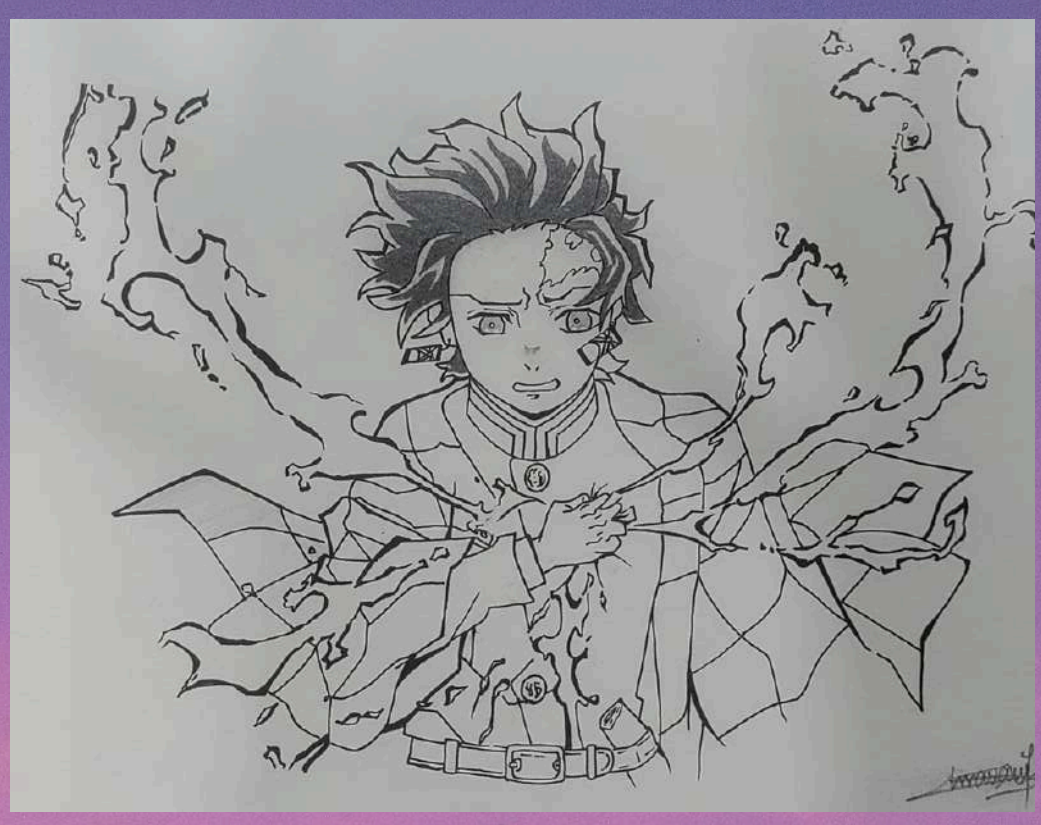


~ Sakshi K.
TE 51

Sketches & Mandala Art







~ Shravani K.
TE 51







~ Sanisa Kamat
TE 51

ZEST

- ATHLETICS
- FOOTBALL/FUTSAL
- CRICKET/BOX CRICKET
- BADMINTON
- KABADDI
- BASKETBALL
- TABLE TENNIS
- CHESS & CARROM
- TUG OF WAR

ATHLETICS



SARA PATHAK TE-2K

100 METER



SARA PATIL TE-6I

200 METER



RELAY (Ist)

AYUSH NANGARE TE-6I

JOESPH EBSION AN-2K

VIVEK MOHITE AN-6I

RUGVED DALVI AN-6I

RELAY (GIRLS)

RELAY 1ST



Sara Patil TE-6I

Hrshita Balakrishnan AN-6I

Niharika Walunj AN-2K

Vibha Shetty AN-4I

RELAY 2ND



Manaal Khan TE-6I

Sara Pathak TE-2K

Pranali Daware AN-4I

Utkarsha Sapkale AE-4I

TABLE TENNIS



SAKSHI KHARADE TE-6I

GIRLS SINGLES (2nd)



SAKSHI KHARADE TE-6I
DISHA GUPTA AN-6I

GIRLS DOUBLES (1st)



PRASANNA NADGAUDA TE-6I
RONIT ANEGUNDI AN-4I

BOYS DOUBLES (1st)



PRASANNA NADGAUDA TE-6I
SAKSHI KHARADE TE-6I

MIXED DOUBLES

BADMINTON



Samruddhi Gadhave
TE-4I

GIRLS SINGLES (1st)



Diya Bhalerao
TE-4I

GIRLS SINGLES (2nd)



Akshata Biradar TE-4I
Pallavi Patil TE-4I

GIRLS DOUBLES (1st)



Sakshi Kharade TE-6I
Shubham Yadav TE-6I

MIXED DOUBLES (2nd)



Meet Dagade TE-2K
Atharva Parthe TE-2K

BOYS DOUBLES (1st)



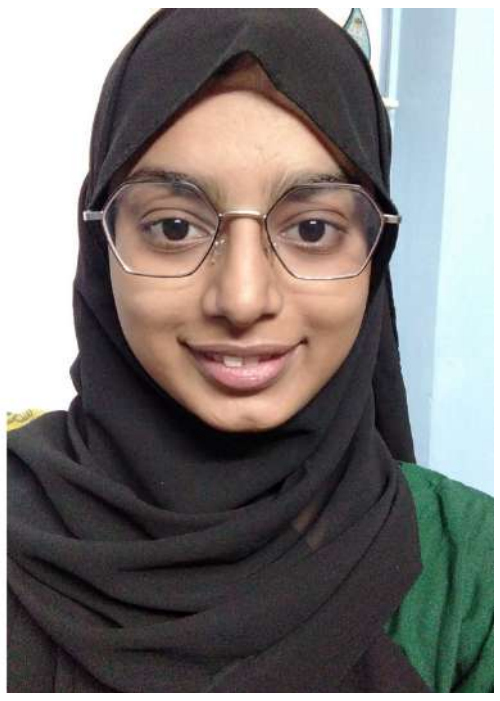
Malhar Jadhav TE-6I
Shubham Yadav TE-6I

BOYS DOUBLES (2nd)

CARROM



DAANIAL MIRZA TE-6I



MUNIRA SHAIKH
TE-6I

BOYS SINGLES (1st)



DAANIAL MIRZA TE-6I
SAAD MULLA TE-6I

GIRLS SINGLES (1st)



MALHAR JADHAV TE-6I
SAMEER SHRIVASTSA TE-6I

BOYS DOUBLES (1st)



MUNIRA SHAIK TE-6I
SAYALI RAJPUT TE-6I

BOYS DOUBLES (2nd)



SUHAANI KADAM TE-6I
DIA BHALLA AN-6I

GIRLS DOUBLES (1st)

GIRLS DOUBLES (2nd)

CARROM



SAAD MULLA TE-6I
MISTI TE-6I

MIXED DOUBLES (1st)



SUHAANI KADAM TE-6I
ARJUN AN-4I

MIXED DOUBLES (2nd)

CHESS



ARYAN SURYGANDH TE-2K

1ST

FUTSAL



2nd

MANAAL KHAN
BHUMI KELKAR
MUNIRA SHAIKH
RACHEL PILLAI
SUHAANI KADAM
SWETA JADHAV
MADHURA KHARAT

BOX CRICKET



1st

SUHAANI KADAM TE-6I
DIA BHALLA AN-6I
MADHURA KHARAT TE-4I
SHRAVANI KATRE TE-4I
PALLAVI PATIL TE-4I
SAYALI RAJPUT TE-6I
SANIKA JADHAV TE-6I

TUG OF WAR



1st

MANAAL KADAM TE-6I
RACHEL TE-6I
SHREYA V TE-4I
DISHA GUPTA AN-6I
KAVYA GADHAVE AN-6I
TANISHA WAGHARE AN-6I
SHARANYA NAKHWA AN-4I
NIHARIKA WALUNJ AN-2K

BASKETBALL

BOYS



1ST

AYUSH NANGARE TE-6I
JAY KATHAYAT TE-6I
RYAN NORMAN AN-6I
PRATHMESH KARANDE TE-4I
JASPAL SINGH TE-4I
SWAPNIL INGLE AN-4I
YASH RAVAL AN-2K

GIRLS



2nd

MANAAL KHAN TE-6I
DIA BHALLA AN-6I
NIHARIKA WALUNJ AN-2K
HARSHITA BALAKRISHNAN AN-6I

CRICKET (BOYS)



1ST

OMKAR WAYAL TE-6I
AYUSH NANGARE TE-6I
RUSHI HIRE AN-6I
TEJAS SAWANT AN-6I
SHUBH SURYAWANSHI AN-6I
JANISH PATIL AN-6I
DUWYANE PINTO AN-6I
RUGVED DALVI AN-6I
ANANT DALVI AN-6I
DEEP DESHMUKH AN-6I
VIVEK MOHITE AN-6I
ARAAJ ATTAR AN-6I
ASHUTOSH KOLI AN-6I
ARJUN DALVI AN-4I
YASH NANGARE AN-2K

FOOTBALL

AN & TE TEAM



2ND

EDWIN
ADITYA
ZIYAAD
AYAAN
ANUJ
VIVEK
DUWAYANE
AZLAAN
ROFEAN
SHUBHAM
RUGVED
SHRAVAN
MALIKT
KISSNA

TECHNOCRATZ

- TECHNICAL QUIZ
- TECHNICAL MIMIC
- TECHNICAL PAPER PRESENTATION

TECHNICAL QUIZ



2nd

Sanika Jadhav TE 6I

Aditi More TE 4I

TECHNICAL MIMIC

1st

Chandrakant Varande TE 4I

Narayanlal Choudhary TE 4I

Malkit Singh Salas TE 4I

Rohan Pednekar TE 4I





2nd

Deepanshu Ghosalkar TE 6I

Divyesh Jadhav TE 6I

Avil D' Souza TE 6I

Krishna Bhojankar TE 6I

TECHNICAL PAPER PRESENTATION



1st

Omkar Wayal TE 6I

Tanishq Parmar TE 6I

2nd

Suhana Patil TE 4I

Smruti Kushalvardhan TE 4I





3rd

Tanaya Deshmukh TE 4I

Sejal Jogale TE 4I

RESONANCE

- SINGING
- DANCING
- OTHER TALENTS
- FASHION SHOW
- RANGOLI MAKING
- POT PAINTING

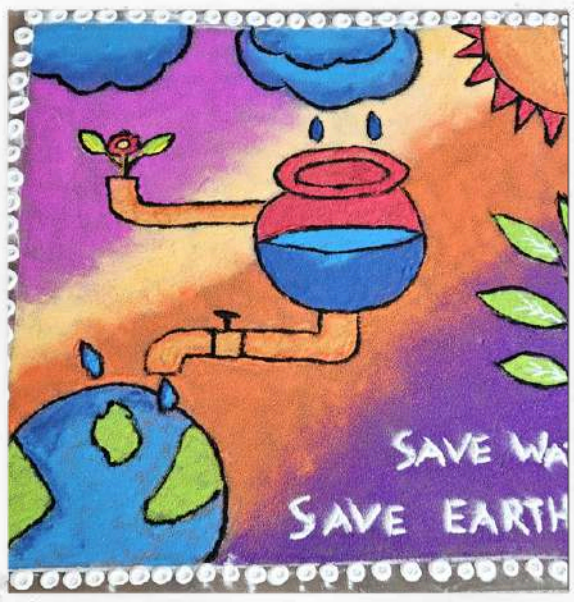











SAVE WA
SAVE EARTH






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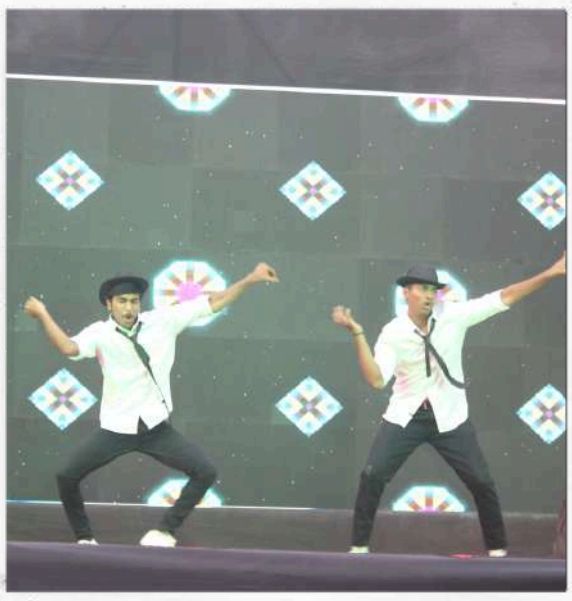



RESONANCE 2024






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JEST 2024
JEST 2024

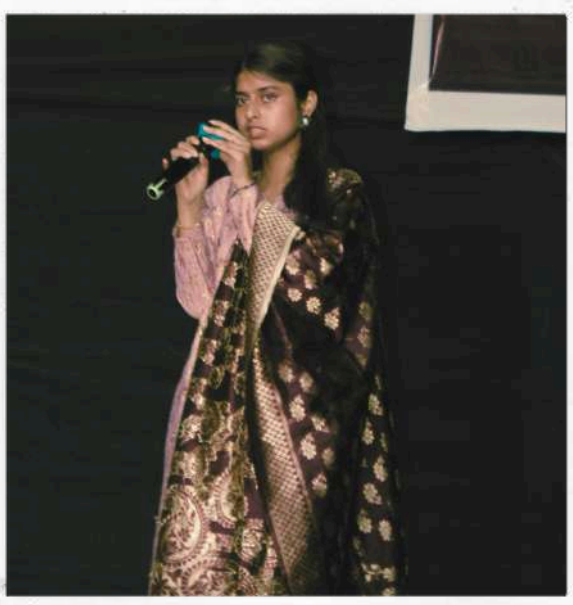



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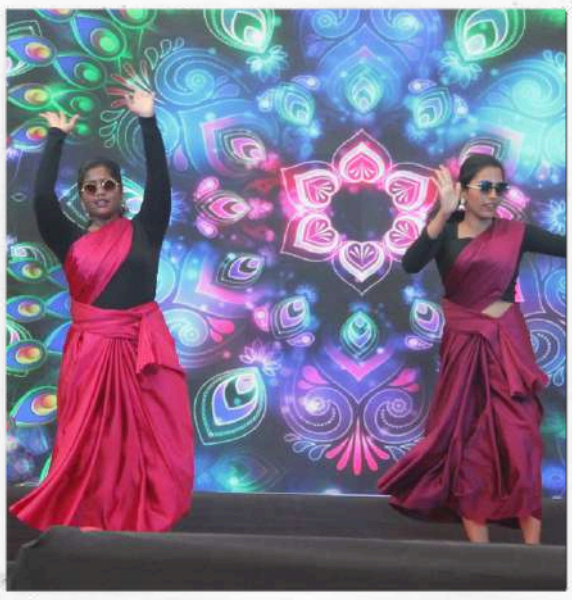
















Legacy Makers

- Third year toppers
- Second year toppers
- First year toppers

THIRD YEAR TOPPERS



Manav Jain
94.47%



Sakshi Kharade
93.76%



Sakshi Sonkul
92.18%

SECOND YEAR TOPPERS



Siddhi Bhoir
91.78%



Sejal Jogale
90.78%



Aditi More
89.56%

FIRST YEAR TOPPERS



PARTH NAIKUDKAR
93.75%



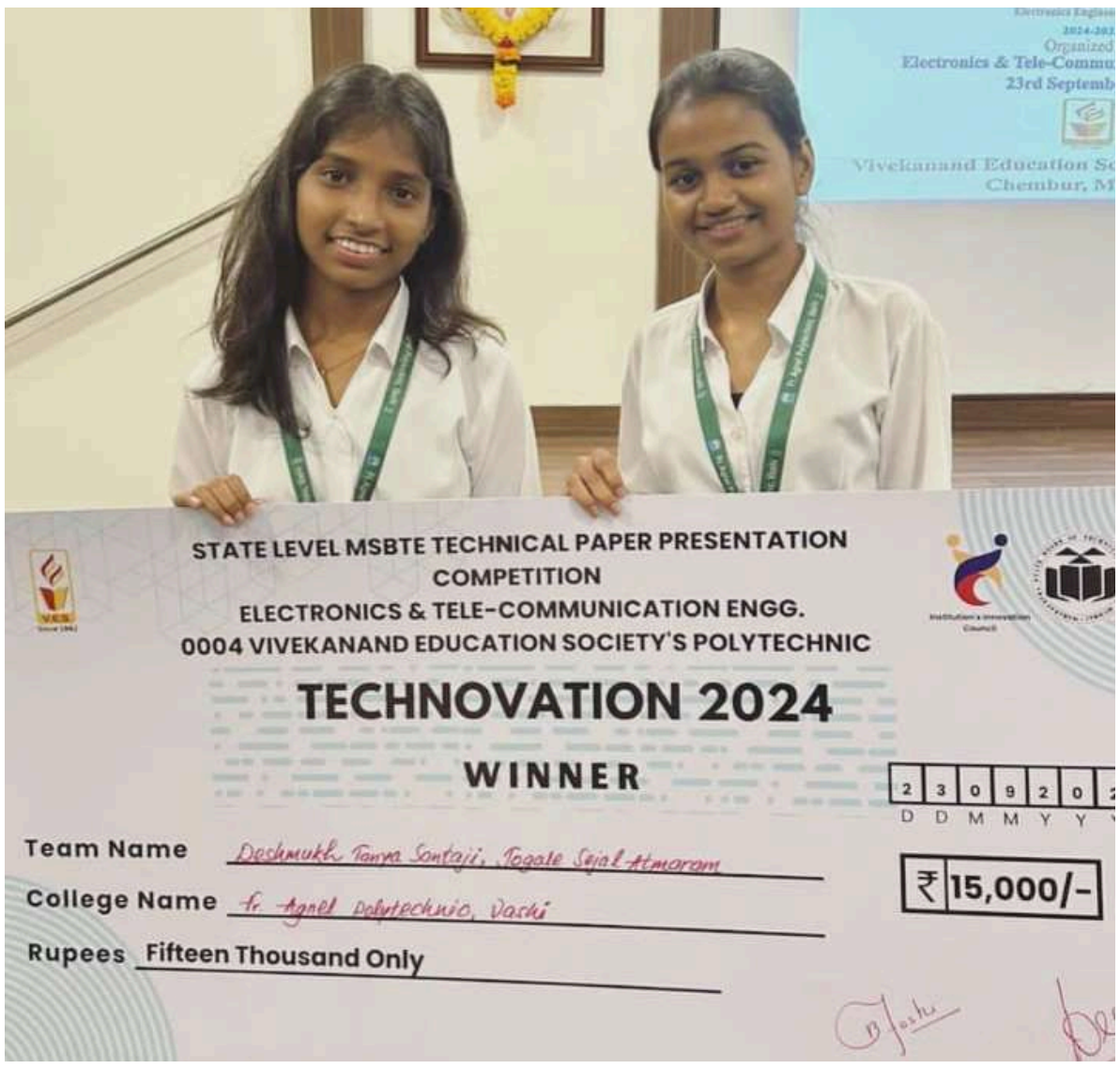
ATHARVA PARTHE
93.50%



ANUJA PATOLE
91%

TECHNICAL PAPER
PRESENTATION
&
QUIZ COMPETITION

State & National Accolades



Sejal Jogale and Tanaya Deshmukh from TE6I have secured 1st prize in State level Technical Paper Presentation held at Swami Vivekananda Institute, Chembur with cash prize of rupees 15000



Sejal Jogale and Tanaya Deshmukh from TE6I have secured 1st prize in National level technical paper presentation held at Bharati Vidhyapeeth Institute, Pune with cash prize of rupees 7000



Stesha Parkhe and Rohan Pednekar from TE 6I won second prize in National Level Paper Presentation Competition held at Bharti Vidyapeeth, Pune.



Parth Naukudkar, Soham Dhanokar and Aayush Bhat from TE/AN 4k have secured 3rd prize in National level technical paper presentation held at Bharati Vidhyapeeth Institute, Pune



Won 1st place at State Level Technical Quiz held at Vidyalankar Polytechnic, Wadala, along with a cash prize of 5000



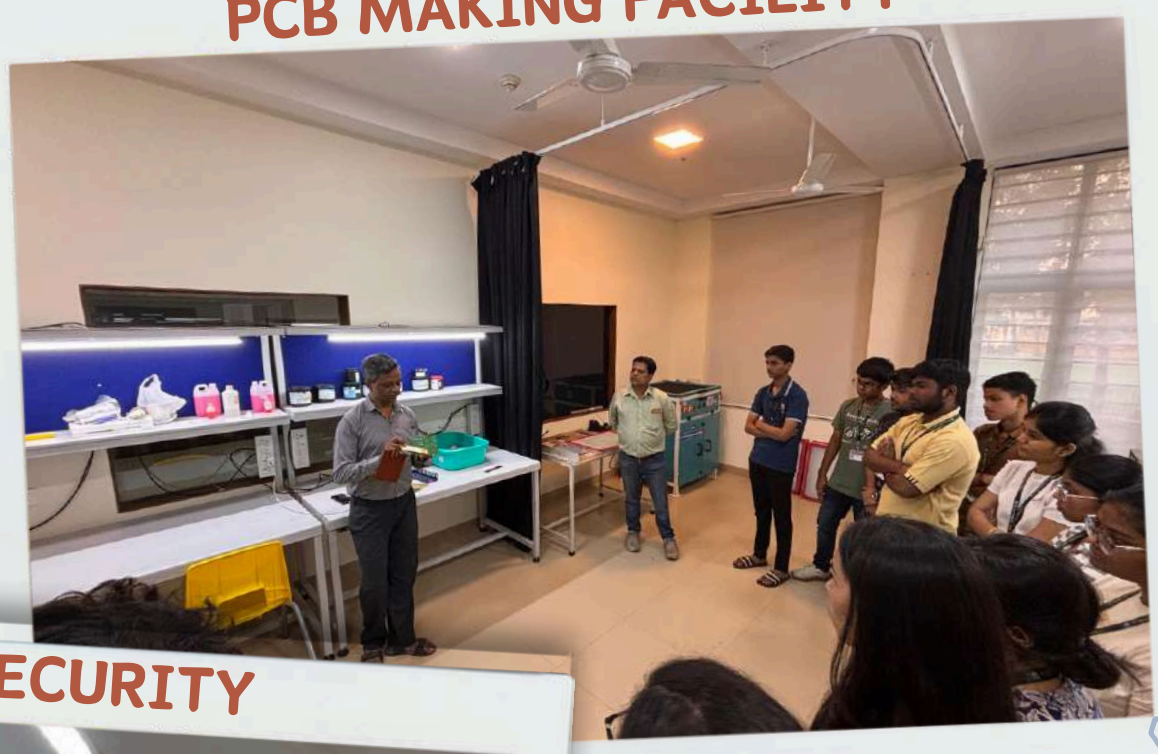
Akshata Biradar & Pallavi Patil of TE 6I second prize in state level Project competition held at Vidyalankar Polytechnic.

Expert lectures and seminars

E-WASTE MANAGEMENT



PCB MAKING FACILITY



CYBER SECURITY



DRONE TECHNOLOGY



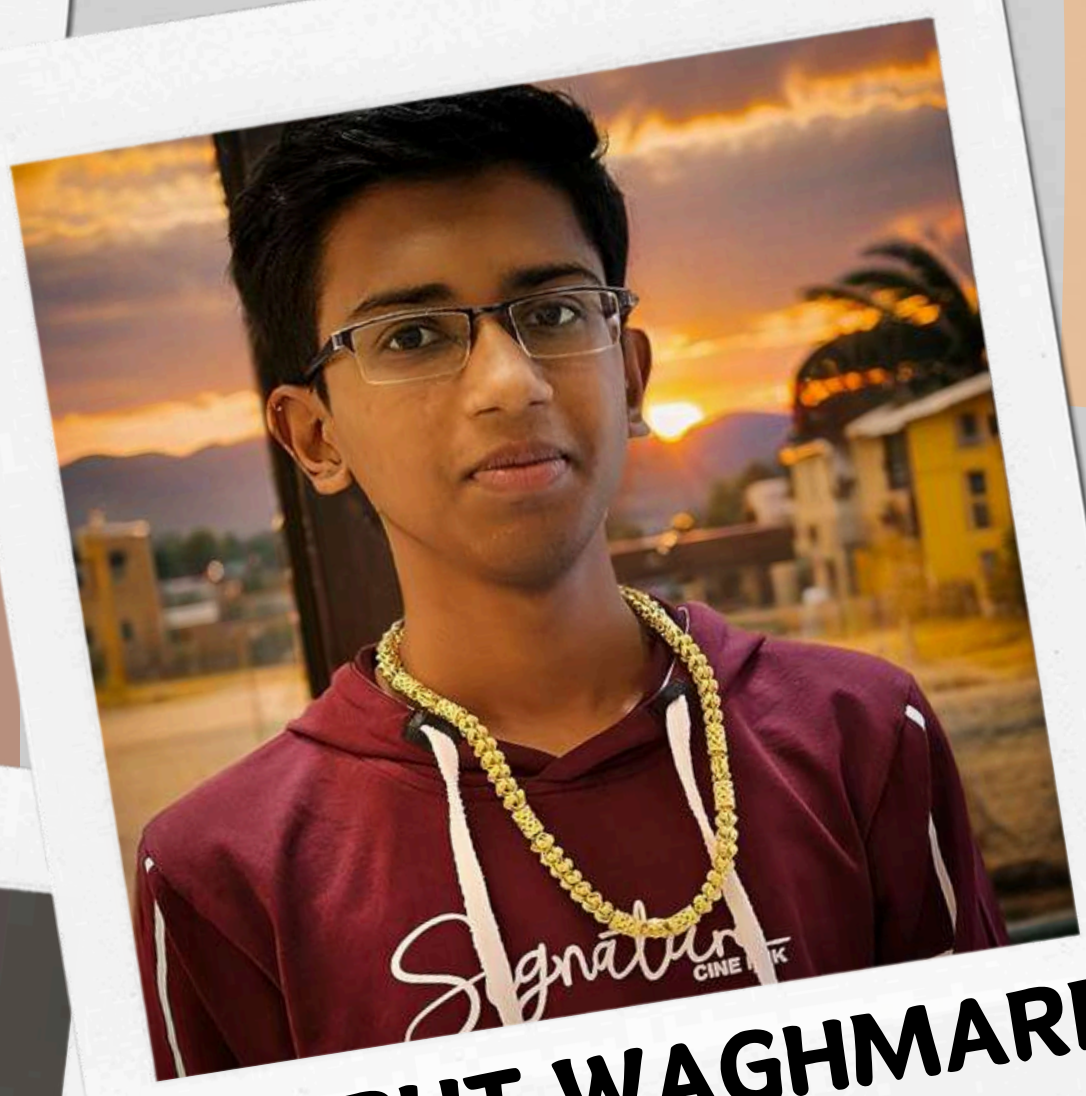
GEN AI



EDITING BOARD



BHAVIKA BIJWASANIA
TE 5I



SUSHRUT WAGHMARE
TE 3K



PARTH NAUKUDKAR
TE 3K

EDITING BOARD

Faculty Coordinator



ARCHANA WASULE



SONALI SHERIGAR